Plan for Project 4

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Dear Peter and Paul,

Based on the recent instructions and the available time frame, we made the following plan and preliminary analysis for our research project:

- **Data Cleaning and Filtering:** We first clean the dataset by removing missing (`NaN`) values and unphysical negative readings. Relative humidity values slightly above 100% during nighttime fog (up to 100.1%) are capped at 100% to account for sensor tolerance, and small nighttime negative shortwave radiation offsets (-3 to 0 W m^{-2}) are treated as zero. Next, we remove North winds (315°–360° and 0°–45° at 8.0 m) to avoid wind distortion caused by the mast itself. Finally, we filter out low wind speeds across all five heights ($0.5, 1.0, 2.0, 4.0$, and 8.0 m). Because a strict threshold of $WS > 2.0$ m s^{-1} at the lowest sensor (0.5 m) removes 58.9% of the data and leaves only one data point on DOY 259 and DOY 264, we use $WS > 1.0$ m s^{-1} (the physical starting threshold of the anemometer) so that every Day of Year (DOY) has enough data for daily analysis.
- **Daily Evolution of Profiles:** Using the cleaned dataset, we analyze the time series of wind speed, temperature, and wind direction across all five measurement heights (0.5 m to 8.0 m).

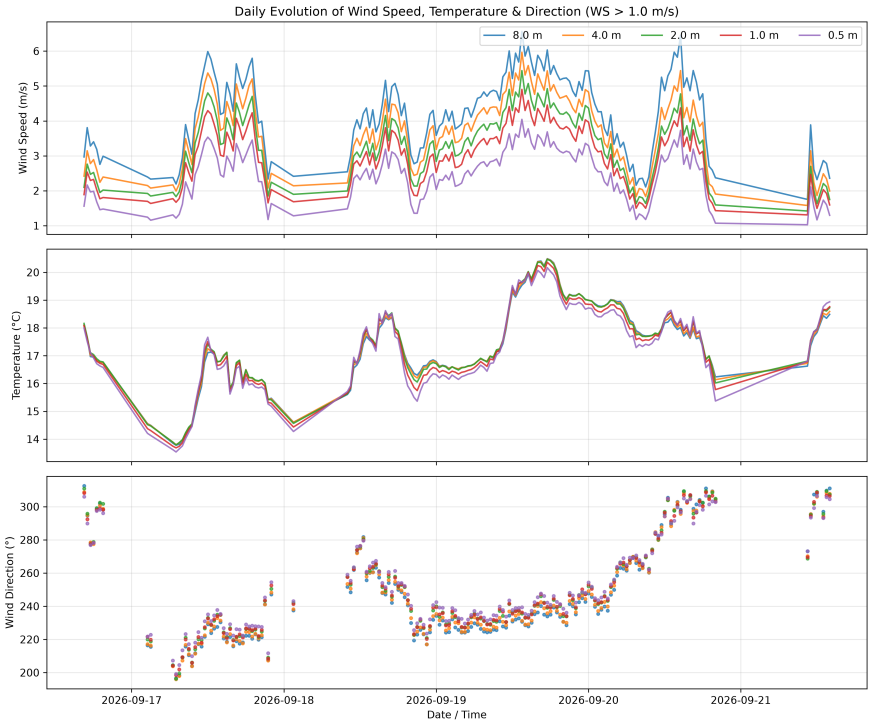


Figure 1: Cleaned daily evolution of wind speed, temperature, and wind direction across the five mast heights ($WS > 1.0$ m s^{-1}, North winds removed).

- **Log-Linearity of Wind Speed and Temperature Profiles:** We check the performance of the vertical profiles by plotting both wind speed and temperature against a logarithmic height scale ($\ln z$). Assuming a zero-plane displacement height of $d = 0$ m over short grass and a von Karman constant of $\kappa = 0.40$, we fit the logarithmic wind profile:

$$u(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right)$$

to find the friction velocity (u_*), roughness length (z_0), and fit quality (R^2) for each individual DOY and for the overall mean. Note that the overall mean curve is weighted toward the windier days because those days have more valid 30-minute data points. We also perform the same logarithmic height fit for the daily and mean temperature profiles.

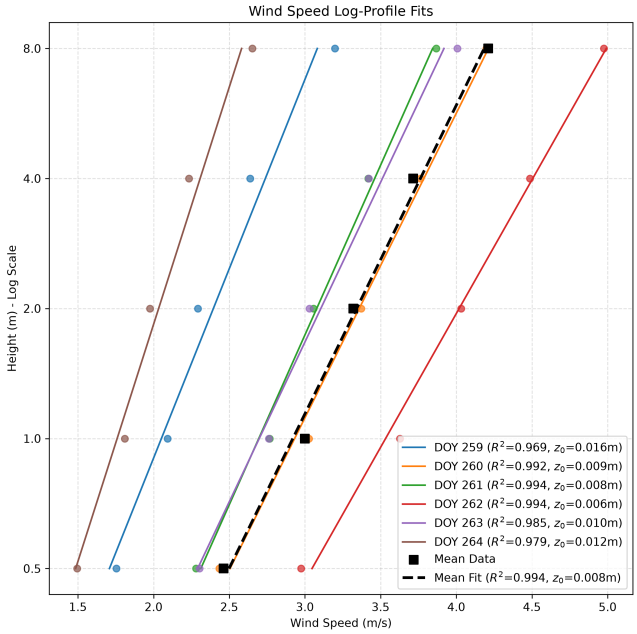


Figure 2: Daily wind speed log-profile fits and overall mean.

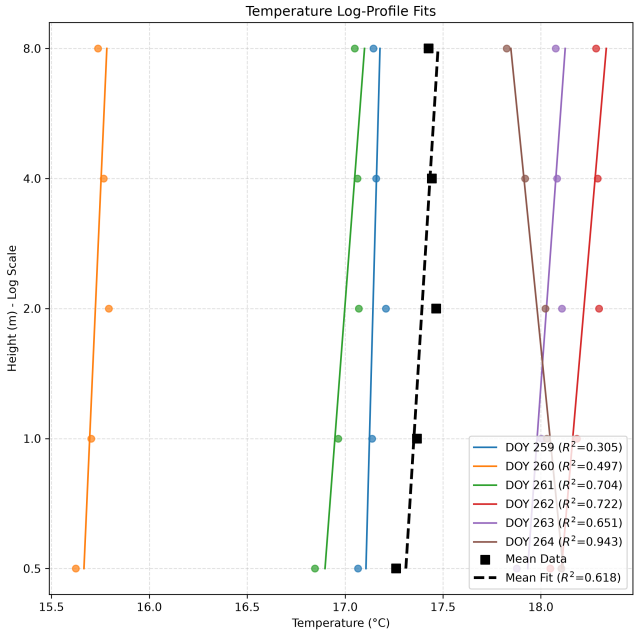


Figure 3: Daily temperature log-profile fits and overall mean.

- **Atmospheric Stability from Temperature Gradient:** We determine the profile relations as a function of atmospheric stability using only the temperature difference between the top and bottom of the mast ($\Delta T = T_{8.0\text{m}} - T_{0.5\text{m}}$):
  - **Unstable ($\Delta T < -0.10\,^\circ$C):** Daytime surface heating where temperature decreases with height.

– **Neutral ($-0.10\,°\mathrm{C} \leq \Delta T \leq +0.10\,°\mathrm{C}$):** Near-zero vertical temperature gradient.

– **Stable ($\Delta T > +0.10\,°\mathrm{C}$):** Nighttime cooling inversion where temperature increases with height.

For each stability group across all days combined, we plot the mean logarithmic wind speed profile and the linear temperature difference profile $T(z) - T(0.5\text{ m})$.

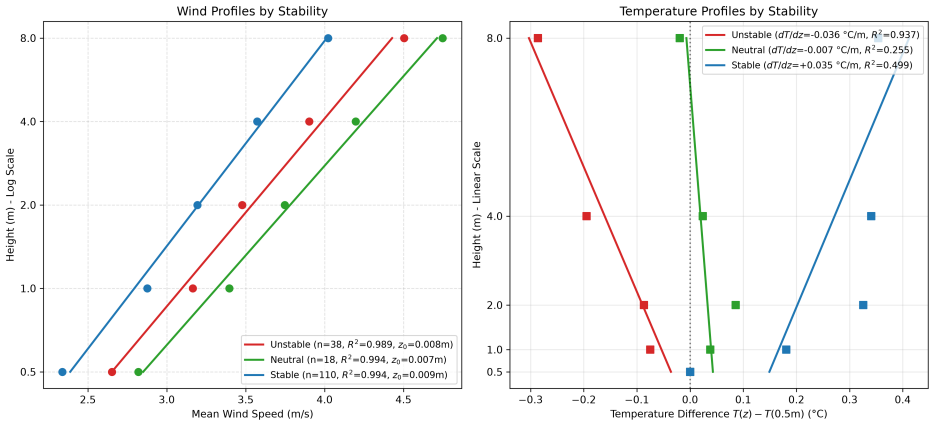


Figure 4: Mean wind speed log-profiles (left) and linear temperature difference profiles $T(z) - T(0.5\text{ m})$ (right) grouped by atmospheric stability.

- **Fetch Analysis:** We group the wind profile data into four directional sectors (East 45°–135°, South 135°–225°, SW–West 225°–270°, and WNW 270°–315°) to see how upwind obstacles at different distances affect the logarithmic fit (R^2) and roughness length (z_0).

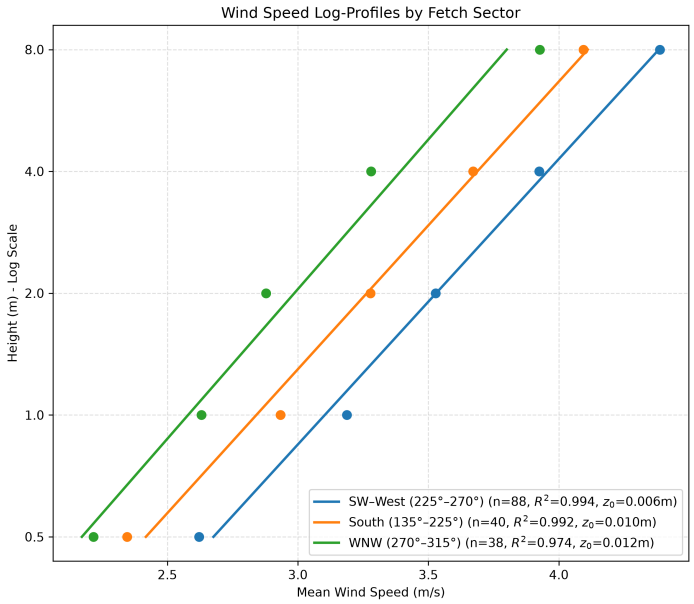


Figure 5: Wind speed log-profile fits grouped by wind direction (fetch) sector.

- **Bulk Fluxes and Sonic Anemometer Comparison:** Finally, we calculate the scaling parameters (u_*, θ_*, q_*) from the differences in wind speed, potential temperature ($\Delta\theta = \Delta T + 0.0098\Delta z$), and specific humidity (Δq) between 0.5 m and 8.0 m. Assuming constant air density ($\rho = 1.20$ kg m^{-3}), specific heat ($c_p = 1005$ J kg^{-1}K^{-1}), and latent heat of vaporization ($L_v = 2.45 \times 10^6$ J kg^{-1}), we compute the sensible heat flux ($H = -\rho c_p u_* \theta_*$) and latent heat flux ($LE = -\rho L_v u_* q_*$) alongside net radiation (Q^*), and we will compare them directly to the turbulence fluxes measured by the sonic anemometer.

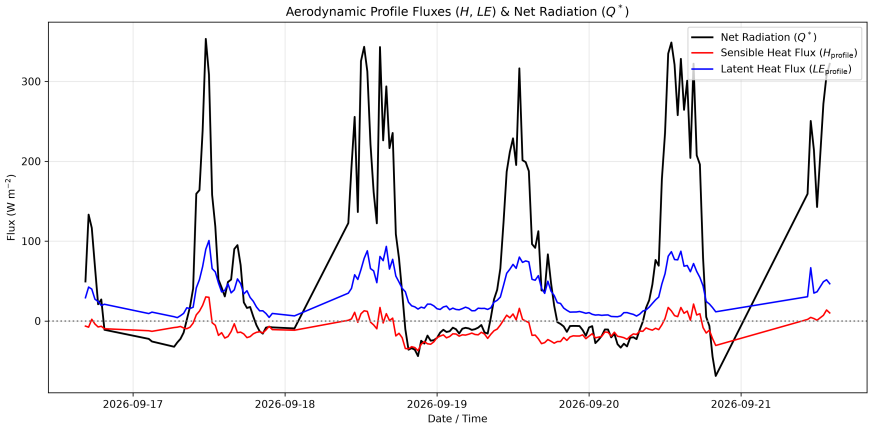


Figure 6: Time series of aerodynamic profile fluxes (H and LE) and net radiation (Q^*).

Best regards,
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